

To following machine learning regression method using r^2

1. Multiple Linear Regression

$$r^2 = 0.9358680970046241$$

2. Support Vector Regression

S.NO	Hyper Parameter	Linear (r value)	RBF (non linear) (r value)	Poly (r value)	Sigmoid (r value)
1	C=0.01	-0.05746	-0.05748	-0.05748	-0.057483
2	C=10	-0.03964	-0.056807	-0.05366	-0.054719
3	C=100	0.10646	-0.05072	-0.01980	-0.030453
4	C=1000	0.78028	0.00676	0.26616	0.1850686
5	C=10000	0.92399	0.371895	0.81296	0.85353
6	C=100000	0.93012	0.7085	0.400210	-0.84337

The regression use R^2 value (linear and hyper parameter($c=100000$)) = 0.93012

3. Decision Tree

s.no	criterion	splitter	R^2 value	
1	squared_error	best	0.9301030	
2	friedman_mse	best	0.913817	
3	absolute_error	best	0.9650880	
4	poisson	best	0.9310756	
5	squared_error	random	0.956625	
6	friedman_mse	random	0.65416	
7	absolute_error	random	0.86235	
8	poisson	random	0.912941	

The Decision tree regression r2 value is (absolute_error, best) = 0.9650880

4.Random Forest

S.No	criterion	n_estimators	R2 Value
1	squared_error	10	0.925277
2	squared_error	50	0.944633
3	squared_error	100	0.946004
4	absolute_error	10	0.928182
5	absolute_error	50	0.940193
6	absolute_error	100	0.945909
7	friedman_mse	10	0.920668
8	friedman_mse	50	0.938895
9	friedman_mse	100	0.941270
10	poisson	10	0.93048
11	poisson	50	0.946354
12	poisson	100	0.941388

The Random Forest regression r2 value is (poisson,50) = 0.946354